PIERFRANCESCO MARTINI

Gran Sasso Science Institute, Viale Luigi Rendina 28, L'Aquila, Italy
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Personal webpage: Link

PERSONAL INFORMATION

Born: November 1999, Avezzano, Italy

Citizenship: Italian

Languages: Italian (native), English (fluent)

RESEARCH AREAS

General Relativity, Analysis of PDEs, Differential Geometry

EDUCATION

Gran Sasso Science Institute (GSSI) November 2023 - Ongoing
Ph.D. in Mathematics *L'Aquila, Italy*
· Advisor: Professor Gabriele Benomio

Sapienza, Università di Roma September 2021 - October 2023
Master Degree in Theoretical Physics *Rome, Italy*
· Final grade: 110/110 cum Laude
· Thesis: *Topological transport in flat bands: a mathematical approach*
· Advisor: Professor Gianluca Panati, co-advisor: Professor Domenico Monaco

Sapienza, Università di Roma September 2018 - October 2021
Bachelor Degree in Physics *Rome, Italy*
· Final grade: 110/110 cum Laude
· Thesis: *Study of the Migdal Effect via photoabsorption in light dark matter direct searches*
· Advisor: Professor Andrea Messina, co-advisor: Dr Giovanni Grilli di Cortona

Liceo Scientifico M. V. Pollione September 2013 - July 2018
Scientific high school diploma *Avezzano, Italy*
· Final grade: 100/100 cum Laude

PUBLICATIONS AND PREPRINTS

· Pierfrancesco Martini. *The Maxwell equations on Myers–Perry spacetimes: The Schwarzschild–Tangherlini case.* In preparation, 2026

TALKS, PRESENTATIONS AND SEMINARS

Reading Seminar at GSSI 18/03/2026
Topics in General Relativity, second edition *L'Aquila, Italy*
· Detailed exposition and comment of the paper "Nonlinear nature of gravitation and gravitational-wave experiments" by D. Christodoulou, intended for an audience already familiar with General Relativity.

Journal Club at the University of Cambridge, DAMTP 10/12/2025
PDEs and General Relativity: an introduction *Cambridge, UK*
· Introduction to the language of PDEs applied to General Relativity.

Reading Group at the University of Cambridge, DPMMS 05/12/2025
The Maxwell equations on Schwarzschild–Tangherlini black holes *Cambridge, UK*
· Overview talk to my ongoing research.

Junior Analysis Seminar at Imperial College London 03/12/2025
The Maxwell equations on Myers–Perry black holes *London, UK*

· Invited research seminar

Reading Seminar at GSSI 03/12/2025
Topics in General Relativity *L'Aquila, Italy*

· Two 90-minute lectures devoted to presenting and discussing the proof of Penrose's Incompleteness Theorem.

TEACHING AND TUTORING

Tutor of the course Fundamentals of Mathematics part II September 2022 – February 2023
Degree course in Chemical Sciences, Sapienza, Università di Roma *Rome, Italy*

One semester of weekly 120-minute tutorial lectures. Activities included reviewing the core concepts covered during the main lectures, presenting worked examples, and providing step-by-step guidance for solving examination-style problems.

Tutor of the courses Geometry and Mathematical Analysis September 2022 – February 2023
Degree course in Civil and Industrial Engineering, Sapienza, Università di Roma *Rome, Italy*

One semester of 5-hour interactive tutoring sessions with small groups of students. Activities included reviewing lecture material, discussing examples, and providing step-by-step guidance for problem solving.

Tutor of the course Physics I March 2022 – September 2022
Degree course in Chemical Sciences, Sapienza, Università di Roma *Rome, Italy*

One semester of weekly 120-minute tutorial lectures. Activities included reviewing the core concepts covered during the main lectures, presenting worked examples, and providing step-by-step guidance for solving examination-style problems.

HONORS

Excellence path of the Master's Degree in Physics September 2021 – October 2023
Sapienza, Università di Roma *Rome, Italy*

Admission to the academic excellence program organized by the Faculty of Mathematical, Physical and Natural Sciences for the Master's Degree in Physics. The program was successfully completed.

Excellence path of the Bachelor's Degree in Physics September 2018 – October 2021
Sapienza, Università di Roma *Rome, Italy*

Admission to the academic excellence program organized by the Faculty of Mathematical, Physical and Natural Sciences for the Bachelor's Degree in Physics. The program was successfully completed.

FURTHER QUALIFICATIONS

Blue Belt in Brazilian Jiu Jitsu 24/06/2026
Awarded by Sensei Mauro Zugaro *L'Aquila, Italy*

Black Belt (I Dan) in Gōjū-ryū Karate 12/12/2016
Awarded by FIJLKAM,
the Italian Federation for Judo, Wrestling, Karate and Martial Arts

REFEREES

- Prof. Gabriele Benomio (Gran Sasso Science Institute)
  Office: PT-13, Zenith Building, Viale Luigi Rendina 26-28, L'Aquila (AQ), 67100, Italy.
  e-mail: gabriele.benomio@gssi.it
- Prof. Rita Teixeira da Costa (University of Cambridge)
  Office: E1.20, Centre for Mathematical Sciences, Wilberforce Road (CB3 0WA), Cambridge, United Kingdom.
  e-mail: rita.t.costa@dpmms.cam.ac.uk

- Prof. Martin Taylor (Imperial College London)
  Office: 677, Huxley Building, South Kensington Campus, London, United Kingdom.
  e-mail: martin.taylor@imperial.ac.uk
- Prof. Gustav Holzegel (University of Münster and Imperial College London)
  Office: 519, Mathematisches Institut, Einsteinstrasse 62, 48149 Münster, Deutschland.
  e-mail: gholzege@uni-muenster.de

Pierfrancesco Martini **L'Aquila, October 28, 2026**

